MAE5005 Computational Fluid Dynamics, Spring 2026

Homework Set 6

Due Monday, April 27, 2026 by 19:00, by online submission through *Blackboard*

17. (30 points) In the lecture, a third approach was proposed to construct a numerical scheme for the advection equation

$$\frac{\partial u}{\partial t} + a\frac{\partial u}{\partial x} = 0. \tag{1}$$

This approach employs "reconstruction, analytical evolution, averaging" - the three-substep approach. If the cell-wise linear reconstruction is used, then the resulting scheme can be written as

$$u_j^{n+1} = u_j^n - \frac{a\Delta t}{\Delta x}\left(u_j^n - u_{j-1}^n\right) - \frac{a\Delta t}{2\Delta x}(\Delta x - a\Delta t)\left(\sigma_j^n - \sigma_{j-1}^n\right), \tag{2}$$

where σ_j^n denotes the slope for the j-cell reconstruction. If the downwind slope $\sigma_j^n = (u_{j+1}^n - u_j^n)/\Delta x$, then the resulting scheme is known as the Lax-Wendroff scheme.

(a) Perform the von Neumann stability analysis for the Lax-Wendroff scheme, and state the stability condition.

(b) Derive the modified equation for the Lax-Wendroff scheme, namely,

$$\frac{\partial u}{\partial t} + a\frac{\partial u}{\partial x} = C_1 a\Delta x\frac{\partial^2 u}{\partial x^2} + C_2 a(\Delta x)^2\frac{\partial^3 u}{\partial x^3} + C_3 a(\Delta x)^3\frac{\partial^4 u}{\partial x^4} + \text{H.O.T.s}, \tag{3}$$

determine C_1, C_2, and C_3. Here H.O.T.s means higher-order terms.

(c) For the same initial condition as stated in Problem 13, $C \equiv a\Delta t/\Delta x = 0.5$ and $N = 160$, study how the L1 and L2 errors vary with time for the Lax-Wendroff scheme.

Note: Some of the analyses are presented in my lecture notes, to be discussed in the class on Monday, April 27.

18. (30 points) Consider again the scheme, Eq. (2). If, on the other hand, the upwind slope $\sigma_j^n = (u_j^n - u_{j-1}^n)/\Delta x$ is used, the resulting scheme is named Scheme B.

Repeat 17(a), (b), (c) for Scheme B.